

TESTS FOR HOOPS MASTER

MCT 231 | Basics of Mechatronics

SECTION 1 — Color Sensor Test & Lifting Mechanism Test

1. Color Sensor Test (Sorting)

Test #	Ball Color	Detected Color	Result
1	Black	BLACK	Pass
2	Black	BLACK	Pass
3	White	WHITE	Pass
4	White	WHITE	Pass
5	Black	Black	Pass

Accuracy: 5 / 5 = __100__%

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SECTION 2 — Shooting Mechanism & Additional Tests

2. Lifting Mechanism Test

Test #	Ball Lifted?	Result
1	Yes	Pass
2	Yes	Pass
3	Yes	Pass

3. Shooting Mechanism Test

Test #	Ball Reached Hoop?	Result
1	Yes	pass
2	Yes	Pass
3	No	Fail

5. IR Sensor Test — Does it correctly detect ball presence / absence? YES

8. Response Time Test — How fast does the system react from ball detection to sorting? 1.5 seconds

9. Repeated Cycle Test — Run the full system 10 times in a row — does it work consistently? YES

10. Power/Electrical Test — Does the system run stable on battery without voltage drops or resets? YES